Fuel switch vs Part load

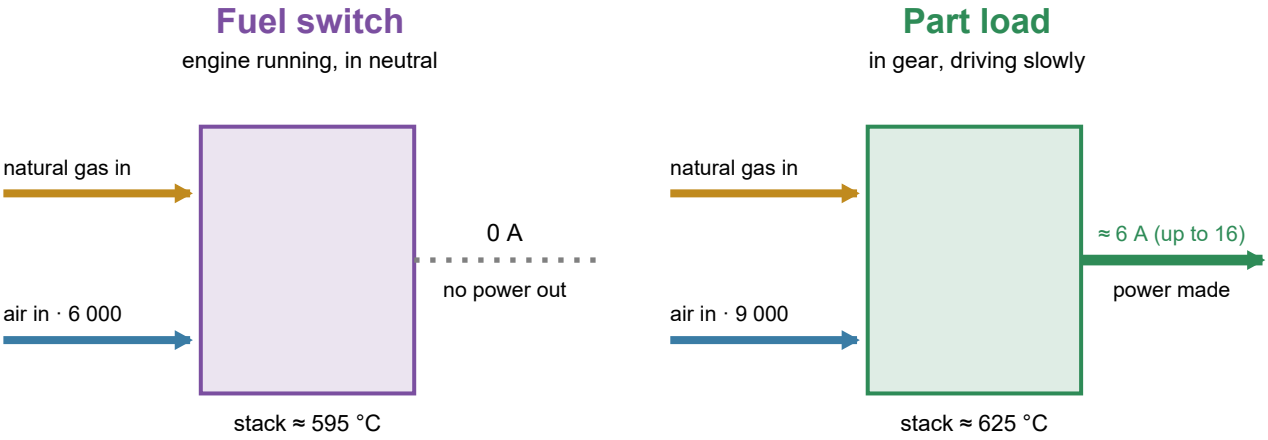
SOFC plant, 8 plant stages · plain-language explainer · 2026-11-04 · interactive version: https://claude.ai/artifact/3U9HtmT5bHmJ5QpajxWdGy

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1. The difference

Both stages burn natural gas in a hot stack. The difference is one thing: in **Fuel switch** no electricity leaves the stack; in **Part load** it does.

Think of a car. Fuel switch is the engine started on its real fuel but left in neutral, idling and warming. Part load is the car in gear, moving, but never at top speed. The plant never reached full (rated) power in our data, so all power-making time is filed as Part load.



Gas in, air in, hot stack: the same in both. Only the power wire on the right changes. Typical (middle) values from the plant log.

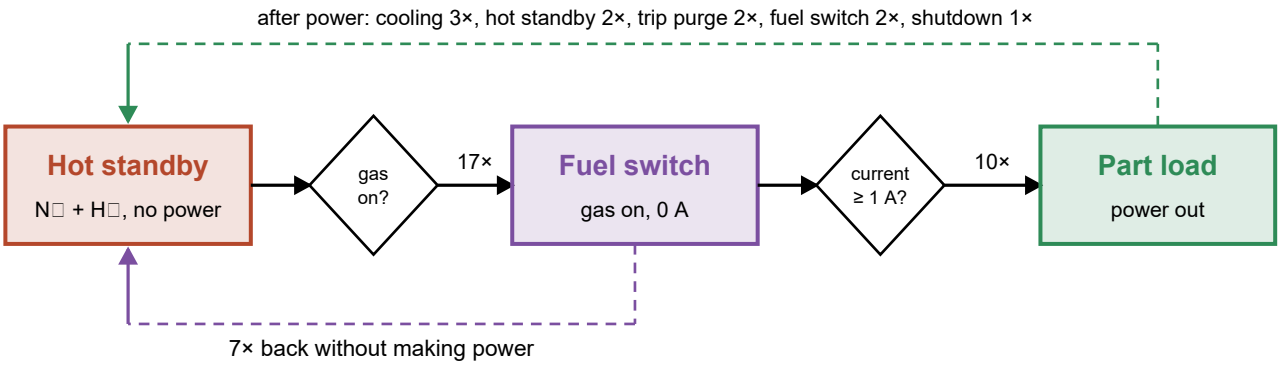
	Fuel switch	Part load
What it is for	Swap the protective gas (nitrogen + hydrogen) for natural gas and get ready	Make electricity
Fuel going in	Natural gas	Natural gas
Electric current out	0 A (always under 1 A)	1–16 A, typically 6 A
Stack temperature	≈ 595 °C, steady	≈ 625 °C, rising while loaded
Cooling air	≈ 6 000 ln/min	≈ 9 000 ln/min
Comes from	Hot standby	Fuel switch only
Time in our data	3.5 h in 19 visits	4.5 h in 10 visits

ln/min = litres of air per minute at standard conditions; A = amperes of stack current. How the code tells them apart: natural gas on and current below 1 A = Fuel switch; control system says Part load (current flowing) = Part load.

If you remember one thing: Same gas, same heat. Fuel switch makes no electricity; Part load does.

2. Where they sit: Fuel switch is the door to Part load

The plant can only reach Part load by passing through Fuel switch. All 10 Part load visits started there. Hot standby keeps the stack hot under a protective gas; operators switch to natural gas (Fuel switch), wait for the stack to settle, then close the power circuit (Part load).



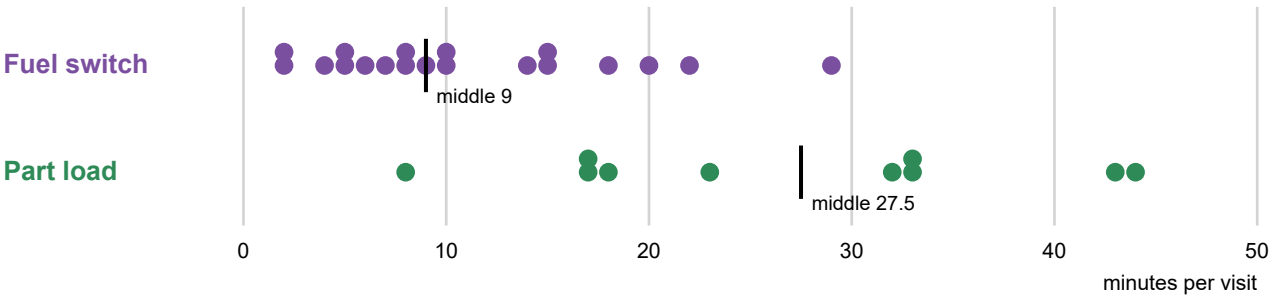
Counts are stage changes in the plant log (Oct 2025 – Mar 2026). Diamonds are the two checks the code makes. Dashed lines are ways back. Other exits from fuel switch: shutdown 1×, trip purge 1×.

In 7 of 17 fuel switches the plant went back to Hot standby without making any power. Seeing natural gas on does not mean power is coming; only the current tells you. A typical wait between gas on and power on was about 10 minutes.

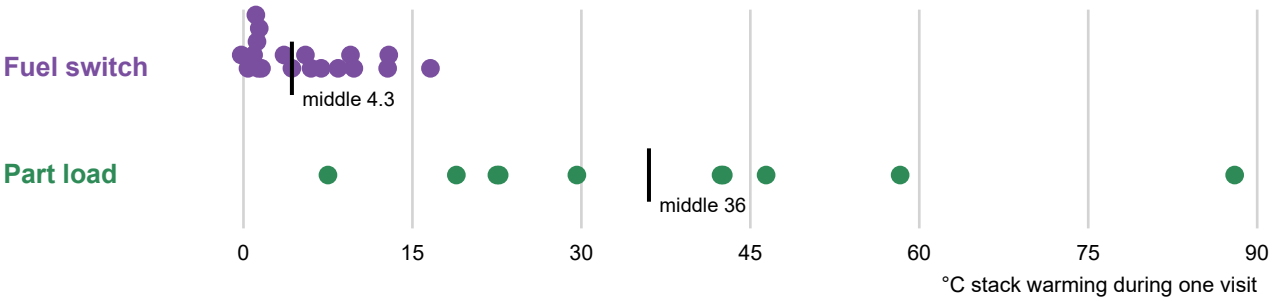
If you remember one thing: Hot standby → Fuel switch → Part load, in that order. Fuel switch can also end with no power at all.

3. What the data shows

Part load visits are longer and they heat the stack. Fuel switch visits are short and the stack barely moves.



Each dot is one visit. Fuel switch: 19 visits, middle 9 min. Part load: 10 visits, middle 28 min.



Making power releases heat inside the stack, so Part load warms it (middle +36 °C); Fuel switch hardly does (middle +4 °C).

Reading	Fuel switch	Part load
Stack temperature (middle, range)	595 °C (577–666)	625 °C (587–677)
Stack current	0 A (max 1.0)	6.4 A (1.0–16.2)
Cooling air	6 000 ln/min	9 000 ln/min
Nitrogen + hydrogen	off	off
What the minutes were	gas on, no load 93 min · ready for load 115 min	load going up 239 min · steady 18 min · going down 11 min
Total time	209 min (3.5 h)	268 min (4.5 h)

More air in Part load carries away the extra heat that making power creates. Both stages are rare: together about 8 hours out of more than 900 hours logged, so any model of them rests on only a few runs.

Source: data_analysis/plant_stages/ built by data_analysis/scripts/plant_stages.py. Rated power never occurred in the log, so it is merged into Part load.

If you remember one thing: Part load lasts about three times longer and warms the stack about eight times more, because it is making power.